

PSTAT 140 Final Project: Microchip Stress-Test Line

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0.1 DEFINE

I am a Quality Control Analyst at a high-volume microchip fabrication facility specializing in semiconductor wafers for consumer electronics. Over the past six months, we've observed inconsistent failure patterns during post-assembly high-temperature stress testing, with defect rates varying by up to 50% between lots and repair costs skyrocketing due to structural microfractures and functional shorts.

0.2 Problem Statement

These defects, ranging from critical calibration errors to minor contamination, are detected through automated thermal cycling and manual optical inspection, yet our current processes lack consistency in detection accuracy and turnaround time. This inconsistency leads to increased rework, delayed shipments, and elevated customer dissatisfaction.

0.3 Goal

Implement robust SPC monitoring and root-cause analysis tools to streamline defect detection, reduce severe defect frequency by at least 30%, and lower total rework and repair costs by 20%, all while maintaining or improving throughput on the stress-test line.

0.4 DMAIC Stages & Tollgates

Below is a concise description of each DMAIC phase tailored to our microchip stress-test scenario, with associated tollgate criteria for advancing to the next phase.

0.5 Define Phase

Purpose: Clarify project scope, objectives, stakeholders, and CTQs. **Activities:**

- Map current process flowchart
- Identify CTQs (e.g., reject rate, cycle time)
- Draft problem statement and project charter

Define Tollgate:

- Project charter signed-off by sponsor
- CTQs validated via initial measurement-system check
- Clear goals and timelines documented

0.6 Measure Phase

Purpose: Quantify current performance and verify data integrity. **Activities:**

- Collect baseline defect and cycle-time data
- Generate SIPOC diagram
- Perform measurement-system analysis (gauge R&R)
- Establish control limits (P-Chart, NP-Chart)

Measure Tollgate:

- Baseline Sigma and defect-rate metrics established
- Measurement system accuracy confirmed
- SIPOC and control charts reviewed with stakeholders

0.7 Analyze Phase

Purpose: Identify root causes of variation and defects. **Activities:**

- Interpret SPC charts for special-cause signals
- Create Fishbone (Ishikawa) diagram
- Construct Pareto chart of defect categories

Analyze Tollgate:

- Root-causes prioritized and validated with data
- Fishbone and Pareto results reviewed
- Key drivers for defects agreed upon

0.8 Improve Phase

Purpose: Develop and test solutions to address root causes. **Activities:**

- Brainstorm improvement ideas (e.g., calibration routines, SOP revisions)
- Pilot proposed changes on sample lots
- Compare before/after defect rates (pilot runs)

Improve Tollgate:

- Pilot results meet or exceed targets (e.g., defect rate $\leq 5\%$)
- Cost-benefit analysis approved
- Implementation plan finalized

0.9 Control Phase

Purpose: Sustain gains and monitor ongoing performance. **Activities:**

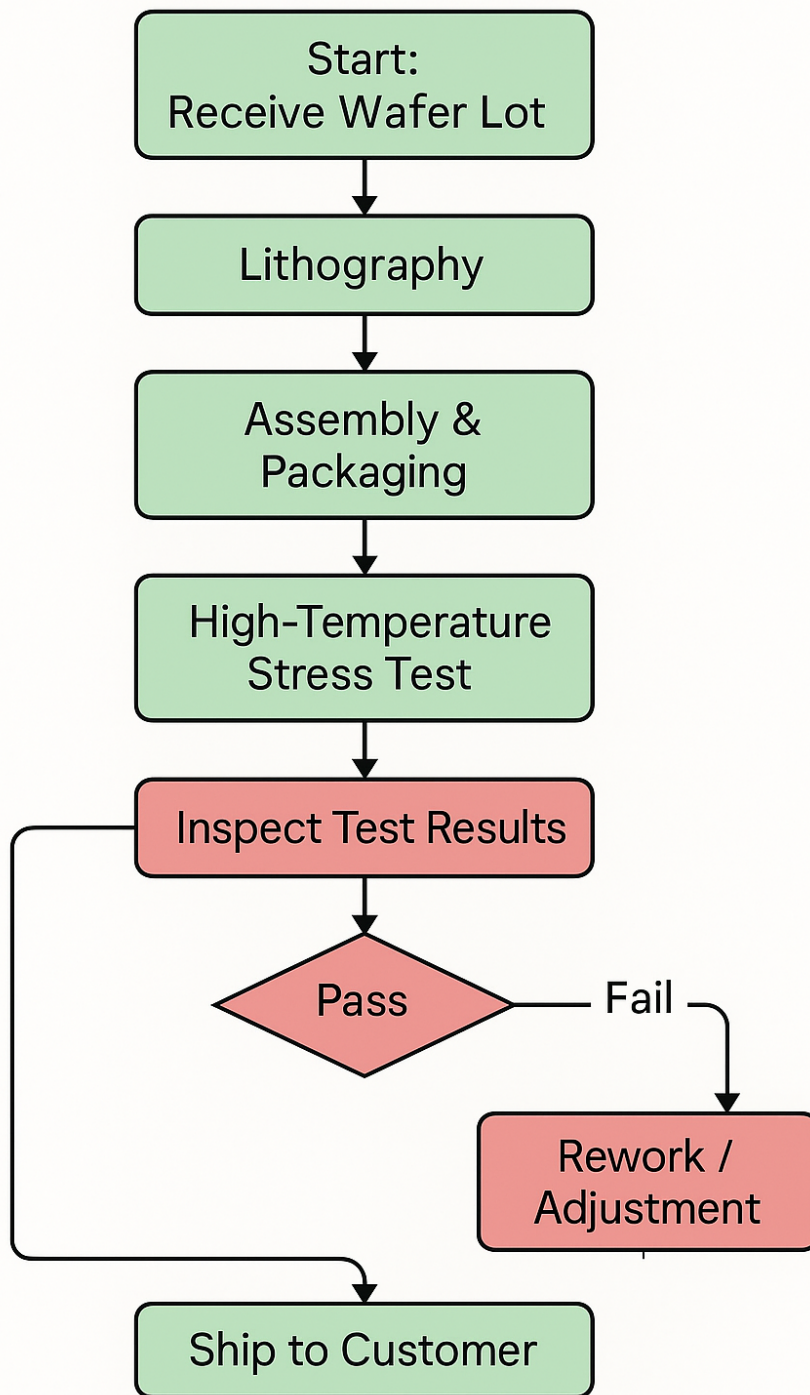
- Update SOPs and training materials
- Deploy real-time SPC dashboards (P-Chart, X-bar)
- Implement Out-of-Control Action Plan and audit schedule

Control Tollgate:

- Control charts demonstrate stable process for ≥ 30 days
- Audits confirm adherence to new procedures
- Transition to process owner for ongoing management

1 (i) Process Flowchart

In this section, we provide a high-level flowchart of the microchip stress-test process, highlighting value-added and non-value-added activities.



2 (ii) Identification of CTQ characteristics

2.1 Project Scope

The scope of this DMAIC project is limited to the post-assembly high-temperature stress-testing step in the microchip-fabrication process. We focus on lots of 80–120 units subjected to a 125 °C thermal shock test followed by inspection.

2.2 Critical-to-Quality (CTQ) Characteristics

- **Reject Rate (p):** Proportion of units failing the stress test in each lot
 - Metric: number of rejects divided by lot size
 - Baseline: historical average reject rate 7%
 - Target: reduce average reject rate to 5%
- **Test Cycle Time:** Time (in minutes) to complete the stress test per lot
 - Baseline: mean cycle time 180 minutes
 - Target: maintain cycle time within $\pm 10\%$ of baseline while reducing defects

2.3 Define Tollgate Criteria

- **Charter Approval:** Sponsor signs off on project scope, objectives, timeline, and CTQs
- **CTQs Validated:** Measurement system analysis confirms that reject rate and cycle time are measurable with acceptable precision
- **Goals Established:** Clear targets for defect rate and test duration are defined and documented

3 (iii) SIPOC Diagram

3.1 Process Map (SixSigma)

3.2 (iii) Measure Phase — SIPOC diagram using the SixSigma package in RStudio

```
library(SixSigma)
# Define basic flow elements
steps      <- c("Lithography", "Etching", "Assembly",
               "High-Temperature Stress Test", "Inspection")
inputs.overall <- c("Wafer Lots", "Test Fixtures", "Stress Test Protocols")
outputs.overall <- c("Approved Chips", "Reject Reports", "Cycle Time Data")

# Define inputs/outputs as empty lists
input.output <- vector("list", length(steps))

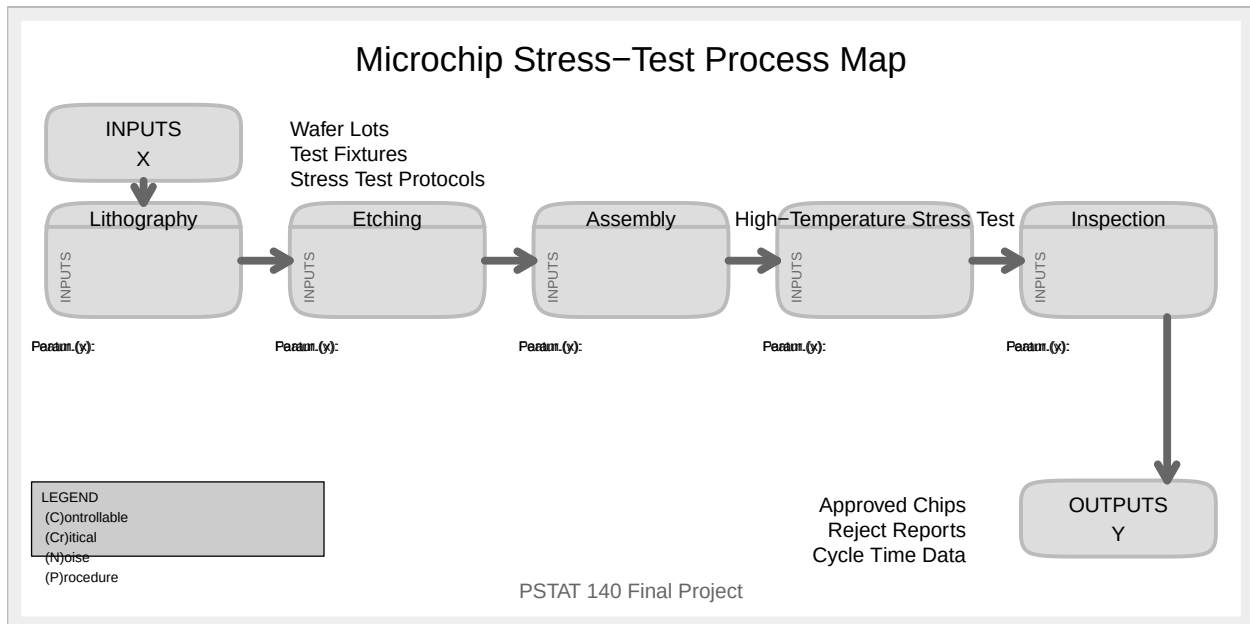
# Empty parameter/feature lists for a clean diagram
x.parameters <- vector("list", length(steps))
```

```

y.features <- vector("list", length(steps))

# Plot simplified map
ss.pMap(
  steps,
  inputs.overall,
  outputs.overall,
  input.output,
  x.parameters,
  y.features,
  main = "Microchip Stress-Test Process Map",
  sub = "PSTAT 140 Final Project"
)

```



(iv) Description of chance and assignable causes Before plotting SPC charts or constructing root-cause diagrams, we distinguish two types of variation in the stress-test process:

- **Chance (Common-Cause) Variation:** Inherent fluctuations due to the normal operation of equipment, material variability, or environmental factors. These are always present but predictable within control limits, for example:
 - Minor fluctuations in chamber temperature ($\pm 2^\circ\text{C}$).
 - Natural variation in wafer thickness across lots.
 - Small operator-to-operator differences in handling time.
- **Assignable (Special-Cause) Variation:** Sudden or unusual events that push the process outside its normal behaviour, requiring immediate investigation, for example:
 - A mis-calibrated chamber heater drifting outside setpoint.
 - A contaminated wafer batch from mishandled transport.
 - A fixture failure causing inconsistent thermal contact.

Recognizing these categories lets us choose the right SPC tool:

- **Common causes** suggest reviewing and tightening control limits or improving overall process capability.
- **Special causes** trigger Out-of-Control Action Plans, fishbone diagrams, and corrective actions focused on the specific anomaly.

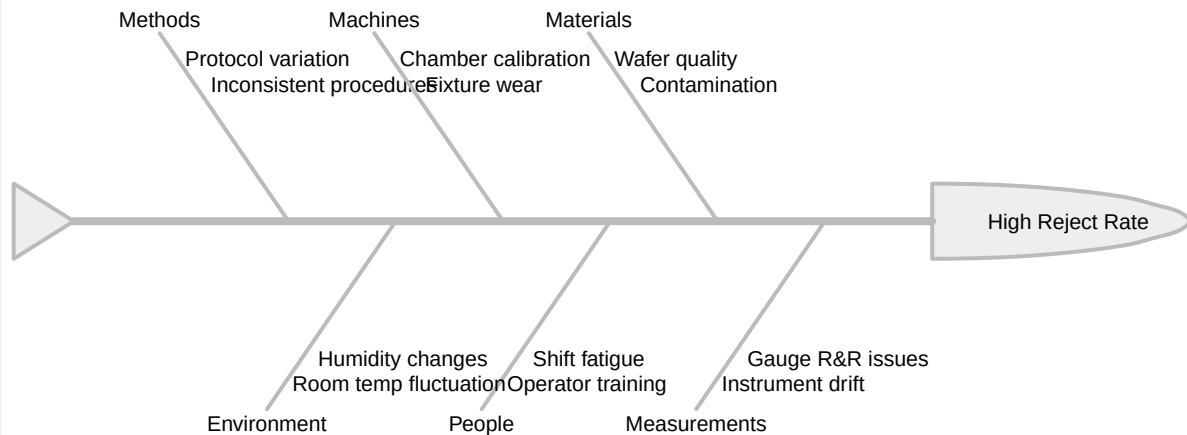
This conceptual framework sets the stage for our P-Charts and NP-Charts in the next section, where we'll detect and respond to out-of-control signals.

```
library(SixSigma)

# Define cause categories and their sub-causes
causes <- list(
  Methods      = c("Protocol variation", "Inconsistent procedures"),
  Machines     = c("Chamber calibration", "Fixture wear"),
  Materials    = c("Wafer quality", "Contamination"),
  Measurements = c("Instrument drift", "Gauge R&R issues"),
  People       = c("Operator training", "Shift fatigue"),
  Environment  = c("Room temp fluctuation", "Humidity changes")
)

# Use ss.ceDiag() to draw the Ishikawa (fishbone) diagram
ss.ceDiag(
  effect      = "High Reject Rate",
  causes.gr  = names(causes),
  causes     = causes,
  main       = "Fishbone Diagram: Potential Causes of Rejects",
  sub        = "PSTAT 140 Final Project"
)
```

Fishbone Diagram: Potential Causes of Rejects



PSTAT 140 Final Project

(v) Out-of-Control Action Plan

Embed your existing Excel-based Out-of-Control Action Plan into the R Markdown with the following chunk. Ensure `out_of_control_action_plan.xlsx` is in your working directory.

```

# Load required package
library(readxl)

# Read the Excel file into a data frame
oocap <- read_excel("out_of_control_action_plan.xlsx")

# Render the table in your report
knitr::kable(
  oocap,
  caption = "Out-of-Control Action Plan",
  booktabs = TRUE,
  align = c("l", "l", "l", "l", "l", "l", "l")
)
  
```

Table 1: Out-of-Control Action Plan

Signal	Immediate Action	Containment	Root-Cause Investigation	Corrective Action	Owner Timeline
Point outside control limits on P-Chart	1. Halt stress-test station immediately				

Signal	Immediate Action	Containment	Root-Cause Investigation	Corrective Action	Owner Timeline
2. Notify Quality Engineer and Supervisor	Quarantine all lots tested since last in-control point	Review calibration logs, operator notes, and environmental data	Re-calibrate chamber; retrain operator on SOPs	Quality Engineer	Within 1 hour
Nonrandom pattern (run/trend) on P-Chart	1. Place chart on hold				
2. Escalate to Process Engineer	Hold future lots pending investigation	Analyze recent process changes, material batch records	Standardize procedure steps; enforce protocol compliance	Process Engineer	Within 2 hours
Out-of-control on Cycle-Time X-bar or R-Chart	1. Pause new test lots				
2. Alert Production Manager	Review pending lot schedule for delays	Inspect test fixture maintenance logs; check resource availability	Service or replace test fixtures; adjust staffing rota	Maintenance Lead	Within 4 hours
Repeated subgroups beyond warning limits (Western Electric Rules)	1. Flag SPC chart				
2. Initiate root-cause checklist	Isolate affected subgroups	Perform Gauge R&R study; verify measurement system integrity	Implement Measurement System corrections; tighten MSA frequency	Quality Analyst	Within 1 shift

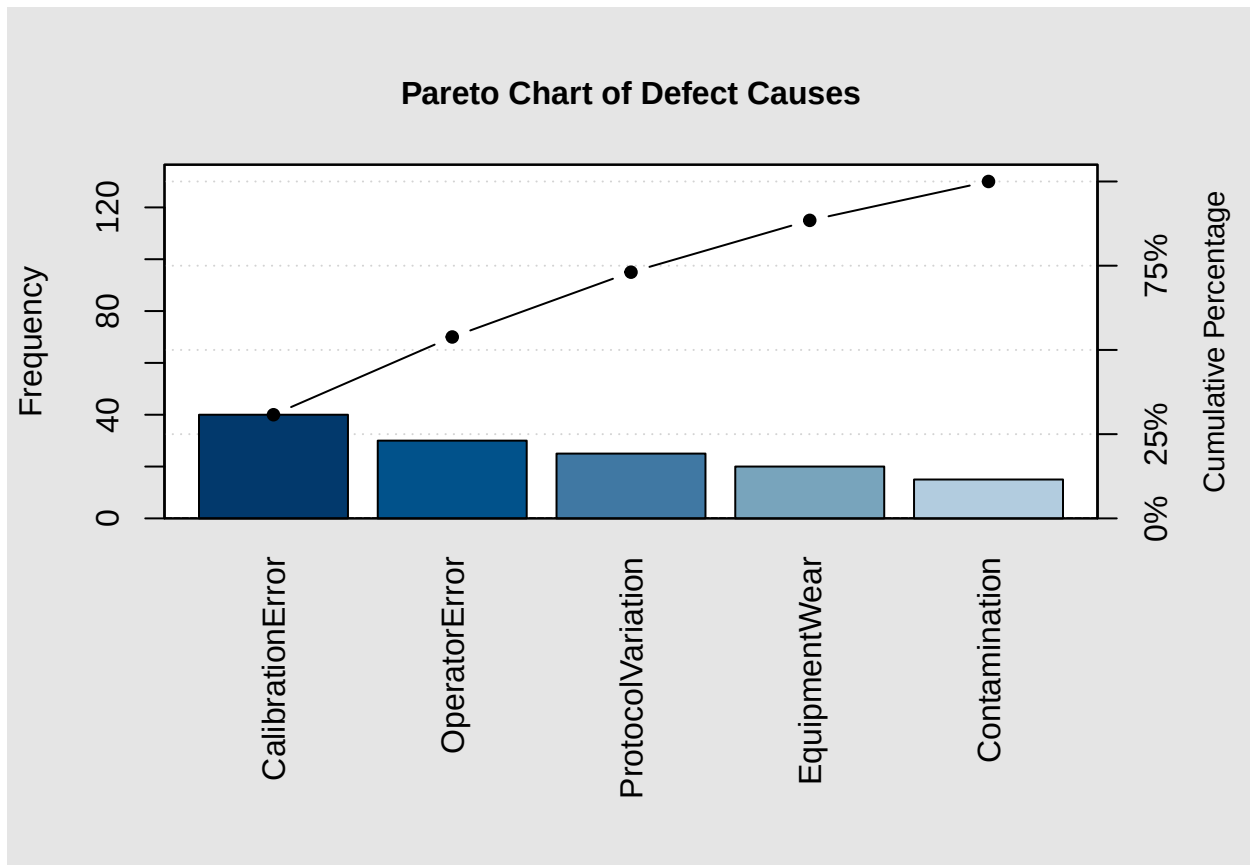
3.3 (vi) Cause-and-Effect (Pareto) Diagram

```
library(qcc)

# Example counts of defect categories
cause_counts <- c(
  ProtocolVariation = 25,
  CalibrationError  = 40,
  Contamination     = 15,
  OperatorError      = 30,
  EquipmentWear      = 20
)
```

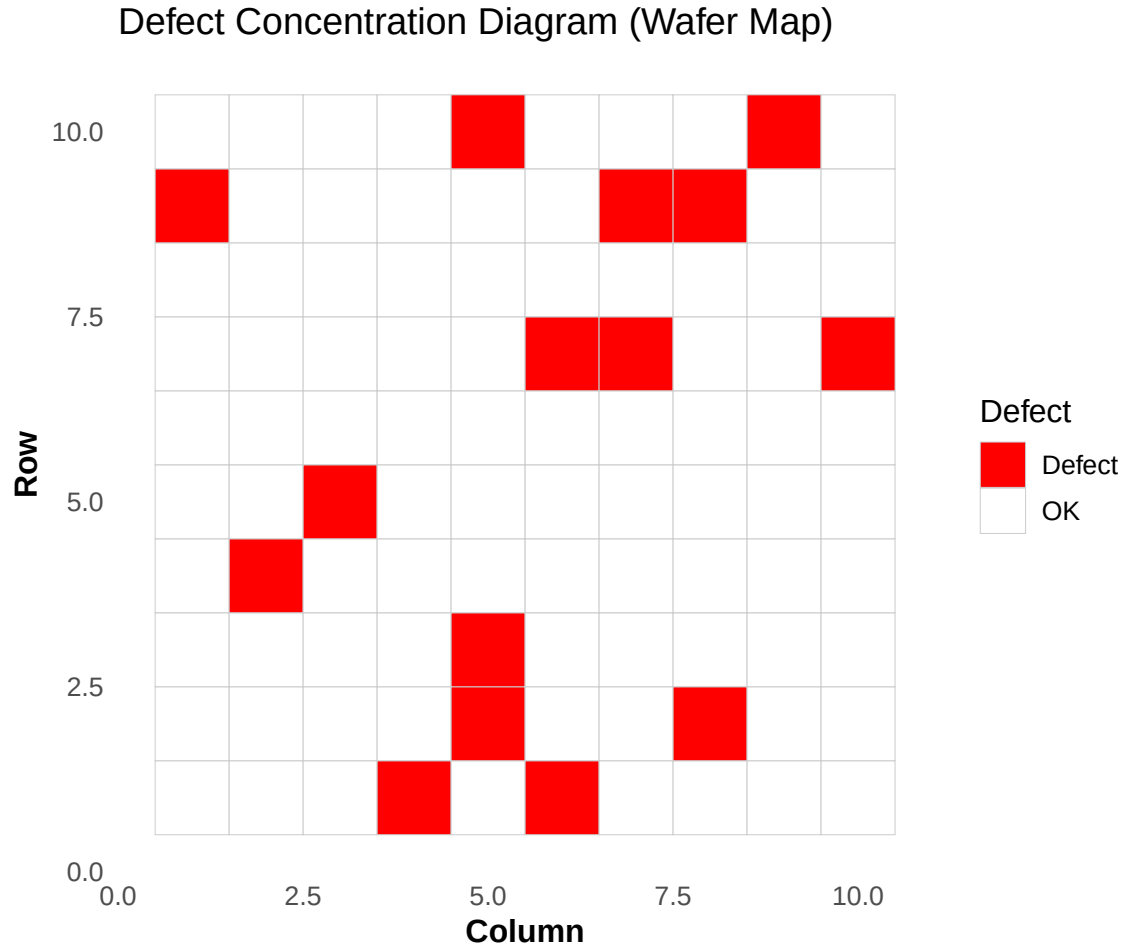
```
)

# Pareto chart: highlights the most frequent causes
pareto.chart(
  cause_counts,
  main = "Pareto Chart of Defect Causes",
  ylab = "Frequency"
)
```



```
##
## Pareto chart analysis for cause_counts
##      Frequency Cum.Freq. Percentage Cum.Percent.
## CalibrationError  40.00000  40.00000   30.76923    30.76923
## OperatorError    30.00000  70.00000   23.07692    53.84615
## ProtocolVariation 25.00000  95.00000   19.23077    73.07692
## EquipmentWear    20.00000 115.00000   15.38462    88.46154
## Contamination    15.00000 130.00000   11.53846   100.00000
```

3.3.1 (vii) Defect Concentration Diagram



(viii)

Check Sheet

A check sheet to record defect occurrences by shift and defect type, facilitating quick tallying in production.

Table 2: Check Sheet: Defect Occurrences by Shift and Type

Shift	Defect Type	Count
Afternoon	Calibration Error	10
Morning	Calibration Error	10
Night	Calibration Error	7
Afternoon	Contamination	7
Morning	Contamination	7
Night	Contamination	11
Afternoon	Operator Error	10
Morning	Operator Error	14

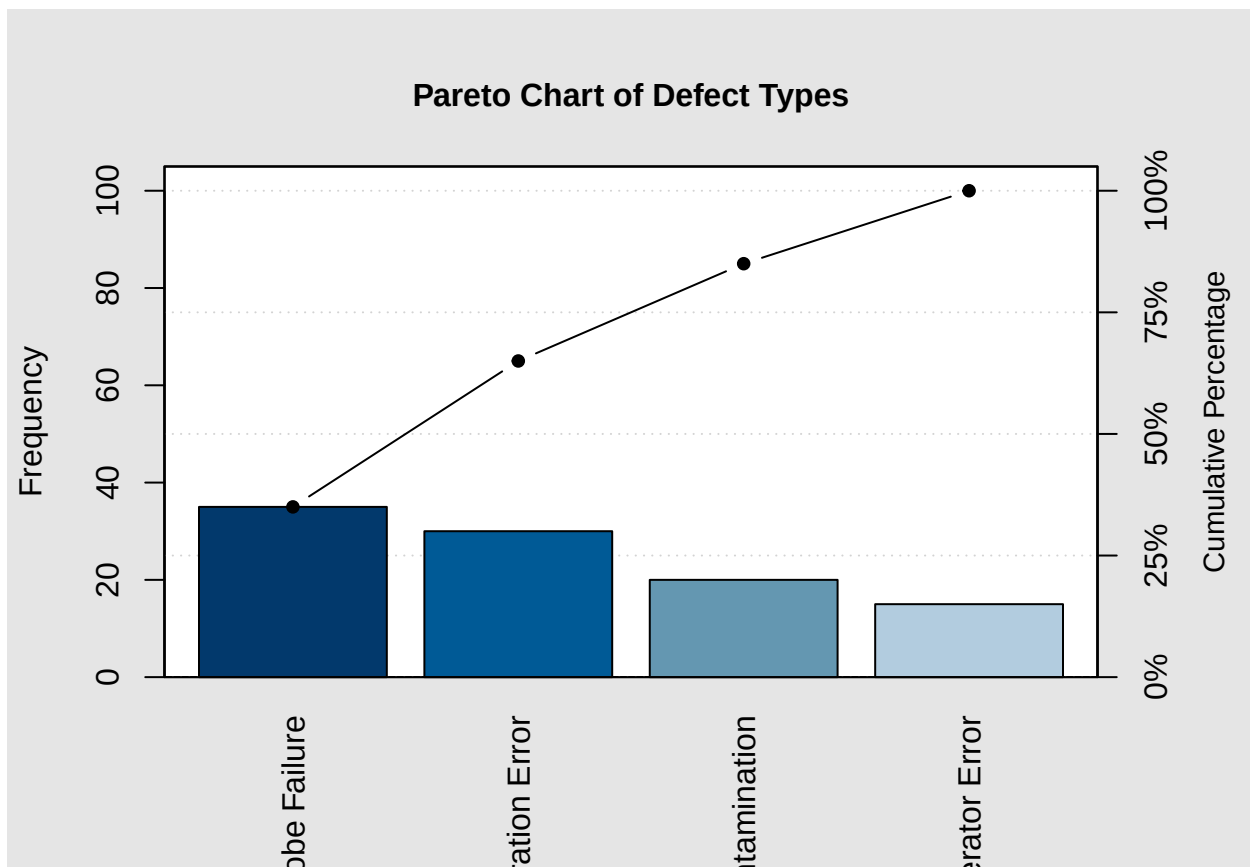
Shift	Defect Type	Count
Night	Operator Error	5
Afternoon	Probe Failure	12
Morning	Probe Failure	13
Night	Probe Failure	14

4 (ix) Pareto Chart in qcc

```
# Load qcc package
library(qcc)

# Simulate defect category counts
categories <- c("Probe Failure", "Calibration Error", "Contamination", "Operator Error")
counts    <- c(20, 35, 15, 30)

# Create a Pareto chart
pareto.chart(
  counts,
  names.arg = categories,
  main = "Pareto Chart of Defect Types",
  ylab = "Frequency"
)
```



```
##
## Pareto chart analysis for counts
##      Frequency Cum.Freq. Percentage Cum.Percent.
##   B          35         35         35         35
##   D          30         65         30         65
##   A          20         85         20         85
##   C          15        100         15        100
```

4.0.1 (x) Value Stream Map Import Code

```
library(readxl)
# Read the VSM data from Excel
vsm <- read_excel("value_stream_map.xlsx")

# Display as a formatted table
knitr::kable(
  vsm,
  caption = "Value Stream Map: Process Steps, VA/NVA, and Bottlenecks",
  booktabs = TRUE
)
```

Table 3: Value Stream Map: Process Steps, VA/NVA, and Bottlenecks

Step	VA/NVA	Cycle Time (min)	Wait Time (min)	Comments / Bottleneck?
Receive Wafer Lot	NVA	5	10	Inspection and logging overhead
Lithography	VA	120	15	Long queue for exposure tool
Etching	VA	30	10	Moderate batch setup time
Assembly & Packaging	VA	45	5	NA
High-Temperature Stress Test	VA	180	20	Chamber capacity is bottleneck
Inspection	NVA	10	15	Manual inspection delays
Ship to Customer	NVA	5	60	Shipping batch consolidation

5 (xi) Demerit System

We establish a demerit system to quantify defect severity across three classes. Each defect type is assigned to a class with a corresponding weight:

Class	Defect	Weight
A	Calibration Error	5
A	Equipment Wear	5
B	Protocol Variation	3
B	Operator Error	3
C	Contamination	1
C	Measurement Drift	1

Justification of Weights:

- **Class A (Weight = 5):** Defects like Calibration Error and Equipment Wear cause critical failures, require extensive rework, and significantly impact throughput and cost.
- **Class B (Weight = 3):** Defects such as Protocol Variation and Operator Error lead to moderate rework and quality issues but are generally correctable with standard procedures.
- **Class C (Weight = 1):** Minor defects like Contamination and Measurement Drift have limited impact on functionality and often fall within acceptable tolerances, incurring minimal rework.

```
# Create demerit system table
demerit <- data.frame(
  Class = c("A", "A", "B", "B", "C", "C"),
  Defect = c("Calibration Error", "Equipment Wear",
             "Protocol Variation", "Operator Error",
             "Contamination", "Measurement Drift"),
  Weight = c(5, 5, 3, 3, 1, 1)
)

# Render the demerit table
knitr::kable(
  demerit,
  caption = "Demerit System: Defect Classes and Weights",
  booktabs = TRUE
)
```

Table 5: Demerit System: Defect Classes and Weights

Class	Defect	Weight
A	Calibration Error	5
A	Equipment Wear	5
B	Protocol Variation	3
B	Operator Error	3
C	Contamination	1
C	Measurement Drift	1

End of DMAIC Report